

Overarm throwing in humans

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Evolutionary aspects

Overarm throwing is a concert of single motions of almost all the muscles and fascia from the tip of the hallux to the tip of the index. Throwing conforms to an individual program: no two people throw in exactly the same way. It is a matter of talent, and even top athlete throwers can enhance their performance by just 10% (Thomas Röhler, personal communication). Throwing also shows the highest gender difference in a motion performance.

Overarm throwing has always been an effective long range weapon to keep predators at distance or injure possible prey. The oldest known javelins are only ~400 000 years old (Thieme, 1997). The so-called Schöninger examples were about 2.20 m long, 3–5 cm in diameter and extremely pointed. They weighed 600–800 g. Experienced javelin throwers can reach distances of up to 100 m with replicas of these oldest known javelins (Milks et al., 2019). Depth of penetration is similar to modern javelins with a spearhead and best when thrown at a distance of 15–25 m. Why do these evolutionary aspects matter? They give us an idea of the relatively short time of change, in evolutionary terms, which occurred to reshape the human body allowing the new action of overarm throwing.

Gender differences in throwing

As overarm throwing has been the basis of hunting in humans and, therefore, was essential for

survival, it is interesting to learn about possible gender differences in this motion. Gender differences in bipedal locomotion (sprinting, marathon running and walking) are non-existent in prepubescent humans and times differ between only 6–10% after puberty (Bhambhani and Singh, 1985). By contrast, the differences in overarm throwing are already significant in boys and girls and increase to 30% in post-pubescent men and women (Young, 2009). Four-year-old boys accelerate a tennis ball up to 42 km/h while girls achieve 20 km/h (-30%). Thirteen-year-old boys show a mean acceleration of 85 km/h while girls of the same age achieve 61 km/h (-30%), with the exception of Aborigine girls (-22%) (Thomas et al., 2010). No other action shows a greater gender difference than overarm throwing.

Biomechanics of overarm throwing

In experienced javelin throwers, it takes around 150 ms from the moment of the contralateral foot touching down before the javelin leaves the fingertips (Valeala, 2012). The javelin is accelerated to more than 100 km/h. The internal rotation of the arm generating the largest impulse lasts only a few milliseconds and is the fastest known action of the human body. The timing for a target-oriented throw is around 3 ms in experienced throwers and 10 ms in those who are less experienced. We distinguish three phases for the throwing action of non-athletes. First, the contralateral foot touches down. During arm-cocking the hip starts rotating, followed by the

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rotation of the shoulder. The arm finally acts like a whip and delivers the object (at the end of the acceleration phase). Figure 33.1A outlines the arm-cocking phase and the acceleration phase of the overhand throw during javelin. Figure 33.1B depicts a series of photographs showing the arm-cocking phase and the acceleration phase from a nearly perfect javelin throw.

Female throwers often start with the ipsilateral foot, which prevents body rotations, and the action mainly comes only from the arm.

In addition to the torso rotation in the waist, anatomical pre-requisites for a successful overarm throw are a close to double shoulder (biglenoidal width) than pelvic width (biac-etabular width), and a reorientation of the shoulder glenoid that points to the side (greater abduction is possible), and which is no longer cranial as in apes. Finally, the humerus has 10–20° lower torsions than in apes. Together, all three components enable greater range and momentum.



Figure 33.1

(A, B) Arm-cocking and acceleration phases of the overhand throw, the relative timing of the cocking motions (A), and the relative occurrence of the opposing acceleration motions (B), adapted from Roach et al. (2013). (C) After run-up, crossover and transfer (C), arm-cocking phase (A) and acceleration phases (B) follow. The nearly perfect throwing of Thomas Röhler, Olympic champion, 2016.

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Torques are generated at each joint from the hallux to the fingertip, accelerating segment masses and creating angular movements. Internal (medial) rotation around the long axis of the humerus makes the greatest contribution to projectile velocity and is the fastest known motion in humans (Hirashima et al., 2007) shoulder internal rotation, elbow extension, and wrist flexion in all speed conditions. The study participants adjusted the angular velocities of these four motions to throw the balls at three different speeds. We also analyzed the dynamics of the 3D multijoint movements using a recently developed method called “nonorthogonal torque decomposition” that can clarify how angular acceleration about a joint coordinate axis (e.g., shoulder internal rotation. However, it is not the result of immediate muscular action, but rather of power amplification due to elastic energy recoil (Roach et al., 2013). According to the authors, the energy storage occurs during the arm-cocking phase, starting with the pre-delivery stride: “The positioning of the shoulder and elbow at this time increases the mass moment of inertia around the long axis of the humerus, causing the forearm and hand to lag behind the accelerating torso. Furthermore, a flexed elbow during the cocking phase enables passive inertial forces to externally counter rotate the arm, stretching the short, parallel tendons, ligaments and elastic components of muscles that cross the shoulder, potentially storing elastic energy in the large aggregate cross-sectional area of these structures. When the biceps deactivate and elbow extension begins, the arm’s moment of inertia is reduced, allowing these stretched elements to recoil, releasing energy and helping to power the extremely rapid internal rotation of the Humerus.” (Roach et al., 2013). Storage of elastic energy is a matter of connective tissue and, in the form of a dense meshwork, fascia in particular.

Soft tissue aspects of throwing

The role of fascial soft tissue in sports performance is underestimated. It is not only a passive tissue with the ability to store elastic energy. Containing contractile cells, free nerve endings and mechanoreceptors, it has an active role in proprioception and mechanics (Wilke et al., 2016).

As mentioned previously, overarm throwing involves torque generation throughout the whole body. Thereby, energy is stored using the stiffness of fascial tissue. Although it appears that fascial tissue should be very stiff, too much stiffness inhibits joint mobility, leading to impairment in the arm-cocking and acceleration phase, resulting in decreased acceleration distance and throwing performance. The torque generation does not happen in every muscle or muscle-soft-tissue complex separately. It is generated and transmitted through myofascial chains (see Chapter 13), involving muscles and soft tissue throughout the whole body (Wilke et al., 2016). Eleven myofascial chains have been described by Myers (2014), based on the principle of continuity of direct linear connection between two muscles. A recent review (Wilke et al., 2016), which addressed evidence-based findings about myofascial chains, showed that not all myofascial chains could be “approved”. Analyzing findings for six myofascial meridians (i.e. superficial backline, superficial frontline, back functional line, front functional line, spiral line and lateral line), only the superficial back line, back functional line and front functional line showed strong evidence. In addition, moderate evidence was found for the spiral line and lateral line. This reveals that fascia and muscles are arranged in a body-wide network with the capability to transmit pain perception, stiffness regulation and torque transmission. With regards to overarm throwing, the functional frontline (M. (musculus) adductor

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longus, M. rectus abdominis, M. pectoralis major) is of special interest (Tittel, 2012). Each transition in the functional frontline (M. adductor longus–M. rectus abdominis and M. rectus abdominis–M. pectoralis major), could be approved by two studies with high consistency (Wilke et al., 2016; 2019). In addition to the functional frontline, the spiral line is also important for overarm throwing. Evidence could only be found for the cranial transitions above the iliac crest (M. obliquus internus abdominis, M. obliquus externus abdominis, M. serratus anterior, Mm. rhomboideus major et minor, M. splenius capitis), but not for the caudal transitions (Wilke et al., 2016). For high performance in overarm throwing, the ability to contract and stretch is very important in those functional systems. Disorders in these myofascial networks can cause functional impairments such as thoracic outlet syndrome, which is a common issue in “throwing athletes”.

Thoracic outlet syndrome in overarm throwing

The thoracic outlet is anatomically defined as the area through which the neural and vascular structures pass from the neck to the armpit. It was first described by Adson and Coffey (1927) as scalenus anticus syndrome. As a kind of entrapment syndrome, thoracic outlet syndrome (TOS) is characterized by a nerve, venous or arterial compression (Archie and Rigberg, 2017). The neurogenic TOS (nTOS) is the most common diagnosis, accounting for 95% of all TOS incidences (Weaver and Lum, 2017). The compression of the brachial plexus can occur in two key sections, the areas above (scalenus aperture) and (more rarely) under the clavicle involving M. pectoralis minor (Sanders and Annest, 2017). The diagnostic for nTOS is subjective in nature, creating controversy about its validity (Weaver and Lum, 2017). The preliminary diagnosis criteria for nTOS

are presence for a minimum of 12 weeks without being satisfactorily explained by other incidences, the symptoms being different to the distribution of a single cervical nerve root or peripheral nerve, and meeting at least one symptom in four out of five categories (Table 33.1).

Further assistive techniques for diagnostics are computed tomography and magnetic resonance imaging to identify bony or fibromuscular abnormalities, which may predispose one to nTOS. Another technique to assess surgical outcomes is scalene injection as a predictive tool. However, it is not clear if there is genetic predisposition to nTOS (Weaver and Lum, 2017). Other forms of TOS, such as venous TOS and arterial TOS entrapments, may be additionally present in patients with nTOS, affecting the arms when in abducted positions (Orlando et al., 2016). In overarm throwing in particular, this is a crucial aspect of diagnostics and has to be examined alongside the symptoms for nTOS.

Therapeutic range from surgical management to medical management

Although medical management has a high success rate with minimal complications, medical management is effective in about 70% of nTOS patients (Weaver and Lum, 2017). As well as for diagnostic purposes, anterior scalene injection is also used as treatment for nTOS, with a better response over a shorter symptom duration (Lee et al., 2011). With regard to athletes with nTOS, full function can be restored in one third by the use of physiotherapy only (Chandra et al., 2014).

Functional therapeutic aspects: scapula alata

In general, being upright against gravity within the first year after birth is related to the release of the upper extremities from the postural support

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Table 33.1 Symptoms of neurogenic thoracic outlet syndrome (nTOS) according to Weaver and Lum (2017)	
Categories	Symptoms
Principal symptoms	(1A) Pain in the neck, upper back, shoulder, arm and/or hand (1B) Numbness, paresthesias and/or weakness in the arm, hand or digit
Symptom characteristics	(2A) Pain/paresthesias/weakness exacerbated with elevated arm positions (2B) Pain/paresthesias/weakness exacerbated with prolonged or repetitive arm/hand use or by prolonged work on a keyboard or other repetitive strain (2C) Pain/paresthesias radiates down the arm from the supraclavicular or infraclavicular space
Clinical history	(3A) Symptoms began after occupational, recreational or accidental injury of the head, neck or upper extremity, including repetitive upper extremity strain or overuse activity (3B) Previous clavicle or first rib fracture or known cervical rib(s) (3C) Previous cervical spine or peripheral nerve surgery without sustained improvement (3D) Previous conservative or surgical treatment for TOS
Physical examination	(4A) Local tenderness on palpation over scalene triangle or subcoracoid space (4B) Arm/hand/digit paresthesias on palpation over scalene triangle or subcoracoid space (4C) Weak handgrip, intrinsic muscles, or digit 5, or thenar/hypothenar atrophy
Provocative maneuvers	(5A) Positive upper limb tension test (5B) Positive 1- or 3-min elevated arm stress test

function used in crawling. This functional process supports the gripping function of the hand and increased upper limb mobility for manipulation. The transition from multiple contact points (e.g. crawling) to two major contact points (bipeds) leads to a reversal of the punctum fixum and the punctum mobile concerning hand and shoulder girdle. For example, in crawling or climbing, punctum fixum is the hand (e.g. touching the ground or the fixation of a grip in climbing) and the shoulder girdle is the punctum mobile (e.g. moving the thoracic region forward). Conversely, with the hand upright for manipulation (e.g. grasping), the shoulder girdle works as punctum fixum to secure the manipulative function of the hand (punctum mobile). The gripping function of the hand must be postural when secured in the shoulder girdle. The shoulder girdle works as punctum fixum. The important basis for this free functioning of the hand is related to the myofascial control of the scapula position.

What does this mean for overarm throwing in humans? For example, as depicted in Figure 33.1, javelin throwing needs efficient torque production “from proximal to distal” (acceleration of the upper extremities while the lower extremities and the trunk are stable) for optimal performance and injury prevention (von Laßberg and Rapp, 2015). This concept was introduced as the “punctum fixum–punctum mobile model”: the “punctum fixum” is defined as the part of the body that is fixed at the rotational axis, and the “punctum mobile” is defined as the free part of the body located most distant from the rotational axis. From a clinical point of view, inadequate myofascial control of the body part that is fixed at the rotational axis (e.g. shoulder girdle and scapula during javelin) is often associated with signs of neurogenic thoracic outlet syndrome. One other clinical sign of inadequate postural security of the scapula is the so-called “scapula alata” (Figure 33.2). The following functional changes are often related to the scapula alata:

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protraction of the shoulder, painful tension of the m. pectoralis major and m. pectoralis minor and the neck muscles, and an anteversion of the head. This malfunctional adaptation is often related to sternosymphysal syndrome (Brügger, 1986). Overarm throwing in humans needs the best possible myofascial functional positioning of the scapula as a pre-requisite for acceleration in overarm throwing.



Figure 33.2 Clinical picture of impingement syndrome of the right shoulder. Note: Arrows mark the insufficient scapular fixation on the right side [scapula alata]

Credit: Christian Puta.

Overarm throwing is characterized by strong one-sided flexors, adductors and interior rotator activity in the acceleration phase, leading to hypertension, for example in the M. pectoralis minor and major. By contrast, extensors, abductors and exterior rotators are functionally inhibited. Figure 33.3 outlines four relevant myofascial and muscular connections (Benninghoff, 2008): (1) craniocaudal connection (M. trapezius pars ascendens–M. levator scapulae); (2) transversal connection (M. serratus anterior pars divergens and superior–M. trapezius pars transversa); (3) upper diagonal connections (M. trapezius pars descendens–M. pectoralis minor); and (4) lower diagonal connection (M. serratus anterior pars convergens–Mm. rhomboidei). As shown in Figure 33.3, hypertension of the M. pectoralis minor causes shoulder depression and reflective tension in the M. trapezius pars descendens. Consequently, the cervical spine is forced into lateral flexion and rotation inhibiting its stabilizing muscles (Mm. scaleni and M. splenius capitis). Because of fast shoulder protraction, high acceleration force in the cervical spine and the associated demand for shoulder and spine stability, muscles like the M. pectoralis minor, M. serratus anterior and Mm. scaleni are key, and are predominantly involved in functional impairments in overarm-throwing athletes (Figure 33.3).

In addition, shoulder stability is affected by M. serratus anterior inhibition. Therefore, it is recommended to implement extensors, abductors and external rotators exercises with a ratio of 2:1 towards their antagonists to prevent functional disorders in the context of overarm throwing in

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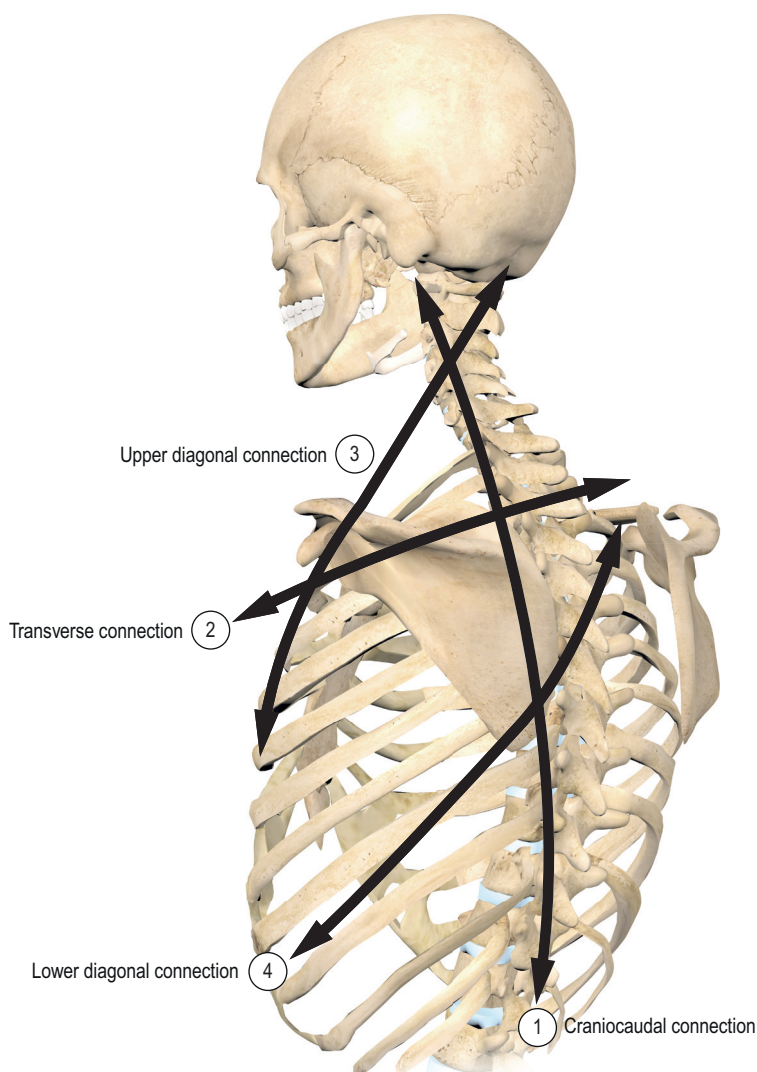


Figure 33.3 Functional muscular connection of the shoulder girdle (Adapted from Benninghoff, 2008)

(1) Craniocaudal connection (M. trapezius pars ascendens–M. levator scapulae); (2) Transversal connection (M. serratus anterior pars divergens and superior–M. trapezius pars transversa); (3) Upper diagonal connections (M. trapezius pars descendens–M. pectoralis minor); (4) Lower diagonal connection (M. serratus anterior pars convergens–Mm. rhomboidei).

humans. Because of its importance to shoulder stability, special attention should be paid to neuromuscular training involving the M. serratus anterior as one functional main “positioner” of the scapula. As mentioned earlier, the functional front

line and the spiral line are involved in overarm throwing by stretching and contraction. To restore myofascial function and prevent or treat TOS, myofascial release techniques are recommended to decrease disorders in those myofascial chains.

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